

MIM-VLA: Learning Physical Interaction Representations from Gripper Motor Feedback

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Abstract—Vision-language-action (VLA) policies infer grasp actions primarily from visual observations and robot state, but do not explicitly represent the physical response observed after contact. We present MIM-VLA, a motor-feedback-based architecture that encodes recent gripper current, position, velocity, and signal validity as a 128-dimensional interaction token τ . A motor-only Motor Interaction Module (MIM) is pretrained with human-reviewed contact and interaction-phase labels and then conditions only the gripper-action pathway of SmolVLA; arm actions and the position-control interface remain unchanged. The same token supports the MEM selector VLM that compares candidate interactions and produces evidence-conditioned selections and explanations. We evaluate MIM-VLA in three real-world settings: comparing the interaction resistance of visually different objects, disambiguating visually similar real and replica objects through active probing, and gently grasping fragile objects, including held-out instances. Across 13 object pairs, MIM-VLA selects the higher-resistance object in 75.0% of trials, compared with 48.8% for the SmolVLA baseline. For the evaluated tasks, the approach uses motor feedback already available from the gripper and does not require an additional tactile array, force-torque sensor, calibrated force estimate, or direct current control.

I. INTRODUCTION

Vision-language-action (VLA) models connect images and natural-language instructions to robot actions [1], [2]. Their visual-language priors support object recognition and task planning, yet appearance does not fully specify how an object will respond after contact. Visually similar objects may have different compliance, while fragile objects of different sizes cannot be safely grasped by replaying one closing position. Joint position alone also cannot distinguish free closing, contact, stable holding, and continued compression.

Force- and tactile-aware policies address this limitation using dedicated sensors or force-aware control [3]–[5]. We instead ask whether actuator telemetry already exposed by a smart gripper can provide useful physical grounding. Motor current is not calibrated force and depends on motion and actuator dynamics; consequently, we model its temporal relation with measured position, velocity, and signal validity rather than applying an instantaneous current threshold.

MIM-VLA, shown in Fig. 2, uses a GRU-based Motor Interaction Module (MIM) to encode recent gripper history as an interaction token $\tau \in \mathbb{R}^{128}$. MIM is pretrained with binary contact and five interaction-phase labels—IDLE, CLOSING, SQUEEZE, HOLD, and OPENING. Video supports label generation but is excluded from the motor-only encoder (Fig. 3). During policy learning, a lightweight projector injects the token only

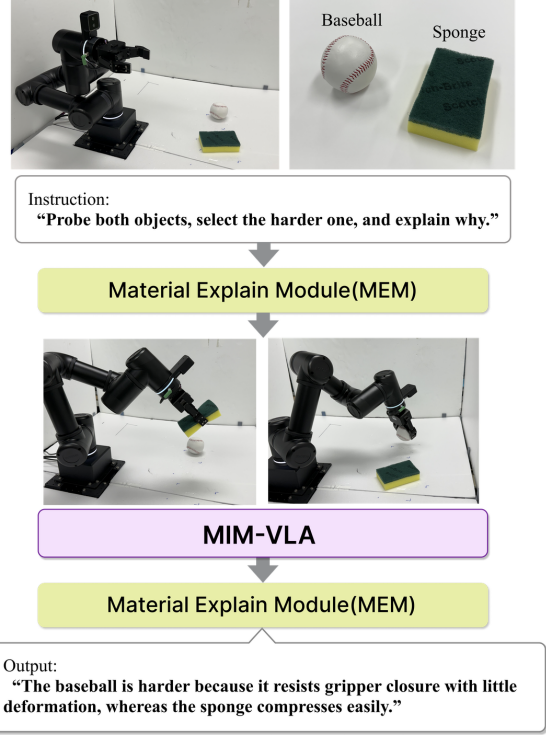


Fig. 1. Interaction-grounded candidate selection. The robot probes both objects, compares their motor responses, selects the object satisfying the requested physical property, and explains its decision.

into a dedicated gripper pathway. The SmolVLM backbone and arm pathway are preserved, and the final outputs remain position commands rather than current or force commands.

For these interaction states, position and velocity describe aperture and closure progression, while current supplies the corresponding motor-load response. MIM remains motor-only to avoid appearance shortcuts; vision stays in the SmolVLM pathway and is not claimed to be replaced by telemetry.

The same representation supports interaction-level reasoning through the Material Explain Module (MEM). Candidate-specific token histories are summarized for a selector VLM, which compares measured interaction responses, selects a candidate, and renders a short explanation; low-level execution remains with the action policy. MEM demonstrates reuse of the same physical token beyond low-level control. We evaluate three capabilities: (i) comparing relative interaction resistance across visually different objects, (ii) probing visually similar

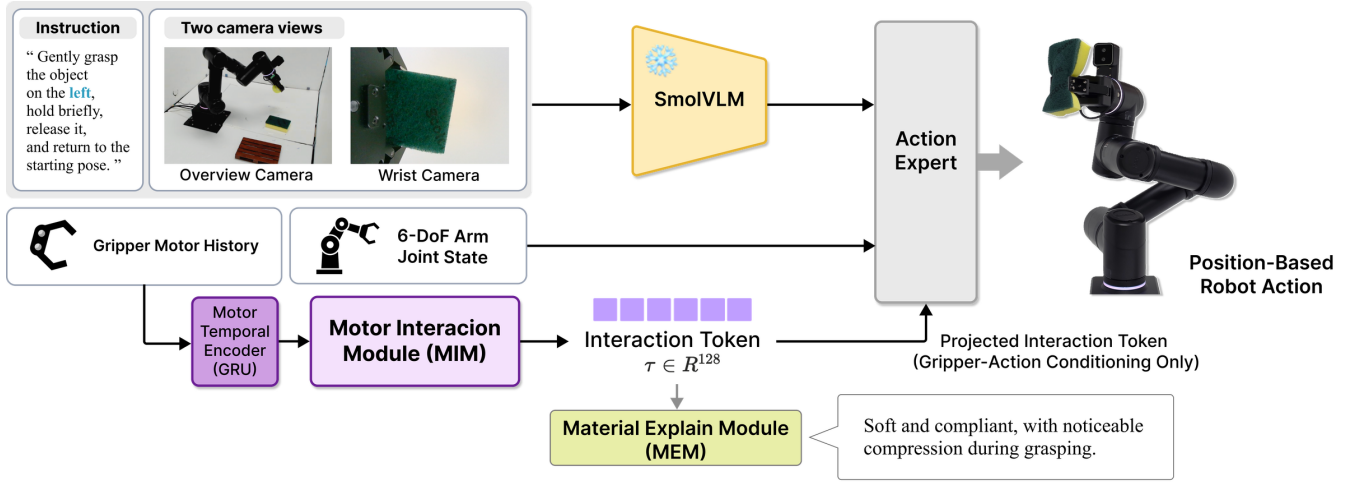


Fig. 2. MIM-VLA architecture. Multi-view observations and language are processed by the frozen SmolVLM backbone. The MIM interaction token conditions gripper-action prediction and supports evidence-conditioned comparison through the Material Explain Module (MEM).

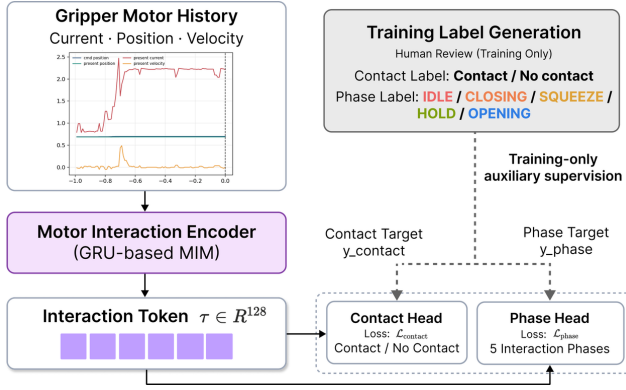


Fig. 3. Motor-only MIM pretraining. Human-reviewed contact and phase targets supervise auxiliary heads during training; video and the commanded position shown for annotation context are visualization only and are excluded from the MIM encoder input.

real and replica objects to identify the more resistant candidate, as illustrated in Fig. 1, and (iii) minimally compressive grasping of seen and held-out fragile objects. Comparisons include vanilla SmolVLA, raw-motor conditioning, and a TA-VLA-inspired temporal-current baseline under the same position-action interface.

Our contributions are:

- a motor-only interaction representation pretrained from temporal gripper telemetry and contact/phase supervision, without calibrated force sensing;
- a gripper-specific integration that preserves the pretrained visual-language backbone, arm pathway, and position-control interface; and
- a shared token interface for closed-loop gripper adaptation and higher-level evidence-conditioned comparison and explanation through MEM, evaluated in three complementary real-world settings.

II. RELATED WORK

A. Vision-Language-Action Models

Vision-language-action models translate pretrained visual-language representations into robot actions. OpenVLA presents an open-source generalist policy pretrained on diverse robot demonstrations and supports efficient adaptation to new settings [1]. More recent VLAs employ continuous generative action models: π_0 adopts flow matching for general robot control [6], while GR00T N1 couples a vision-language module with a diffusion-transformer action module for generalist humanoid manipulation [7]. FLARE augments generative policy learning with future latent representation alignment, encouraging the policy to model consequences in a control-relevant feature space [8]. SmolVLA combines a compact flow-matching architecture with asynchronous inference for efficient deployment on affordable hardware [2]. These methods learn rich representations for action prediction, but they do not explicitly preserve object-associated evidence acquired through prior physical probing. Building on SmolVLA, our approach summarizes gripper motor-response histories as interaction tokens that can be associated with explored objects and used to condition subsequent gripper control.

B. Physical Interaction for Robot Manipulation

Recent work incorporates physical feedback at different levels of the perception-action hierarchy. Among the approaches most closely related to ours, ForceVLA fuses real-time force-torque measurements with vision-language features through a force-aware mixture-of-experts during action decoding [3]. TA-VLA studies torque-aware adapters, temporal torque history, and auxiliary torque prediction for integrating physical feedback into existing VLA architectures [9]. ForceVLA2 introduces force prompts for task-level reasoning and combines them with hybrid force-position control [10]. Tactile approaches align touch with vision and language for contact-rich insertion, as in VTLA [11], or use

tactile feedback at both the planning and action-refinement levels, as in VLA-Touch [12]. Bi-ACT instead learns position, velocity, and force behavior from demonstrations collected with bilateral control [5].

These approaches primarily use explicit force or tactile signals to improve ongoing planning and control, or require a force-aware demonstration and execution interface. In contrast, MIM operates on the temporal current, position, and velocity signals already exposed by the gripper. Wrist video is used only to generate and review auxiliary pretraining labels; it is not an input to the motor-only MIM. We do not interpret current as calibrated force. Instead, MIM encodes the joint evolution of motor load and finger motion into an interaction token, allowing the original position-action interface to be retained.

C. Interaction Grounding with Vision-Language Models

Vision-language models connect visual observations with linguistic descriptions. Qwen2.5-VL supports spatial object localization and temporal event localization in long videos [13], while SmolVLM provides a compact architecture for image and video understanding [14]. Visual evidence alone, however, may not reveal the physical response produced during grasping. PaLM-E demonstrates that continuous robot-state estimates can be interleaved with vision and language to support embodied reasoning [15]. Other work uses VLMs to reason over candidate robot behavior: RoboMonkey samples actions from a VLA and applies a VLM-based verifier to select among them [16], whereas Uncertainty-Aware Policy Steering chooses whether to execute, request clarification, or obtain a corrective intervention according to calibrated semantic and action uncertainty [17].

These systems do not retain candidate-specific motor evidence from the robot’s preceding contact with an object for later comparison. Our Material Explain Module (MEM) summarizes the MIM interaction histories of two probed candidates and provides them to a selector VLM together with the task instruction. The VLM predicts the selected candidate and relative position/current responses, and generates a short explanation conditioned on the same summaries, while the action policy remains responsible for low-level position commands. We evaluate candidate selection in real-robot trials and analyze the effect of interaction- summary capacity using structured response prediction, explanation consistency, and candidate-swap consistency.

III. METHOD

A. Overview

MIM-VLA separates physical-interaction representation from task-level vision and action generation. First, the Motor Interaction Module (MIM) encodes a causal window of measured gripper telemetry as $\tau \in \mathbb{R}^{128}$. Second, the Material Explain Module (MEM) summarizes candidate-specific token histories for a selector VLM that compares prior interactions and produces a structured choice and explanation. Third, the current interaction token conditions a gripper-specific

SmolVLA pathway [2] while the original visual-language and arm pathways are preserved.

Training follows the same separation. MIM is pretrained with contact and phase supervision. The interaction-memory summarizer and selector are then trained with MIM fixed. For downstream robot control, MIM is initialized from the pretrained encoder and jointly fine-tuned with the action pathways, while its auxiliary classification heads remain frozen.

B. Motor Interaction Module

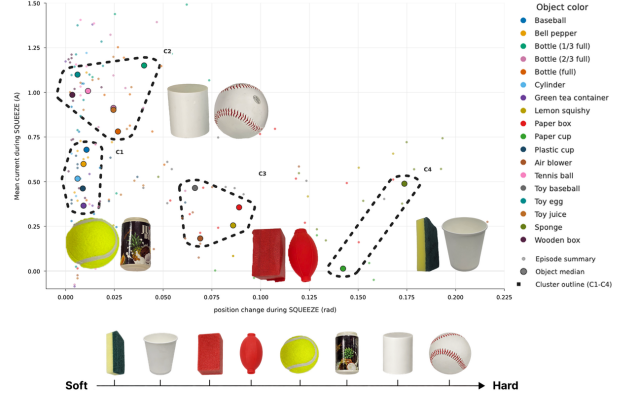


Fig. 4. Qualitative motor-response validation on the MIM pretraining data. The 18 legend entries correspond to the objects used for MIM pretraining. Small points summarize individual episodes, large markers denote object-level medians, and dashed contours indicate recurring response regions (C1–C4) in gripper-position change and mean current during SQUEEZE. Objects with similar interaction resistance exhibit nearby response distributions despite differences in appearance.

MIM processes measured gripper position, velocity, and current together with motion, torque-enable, validity, and timing signals. Commanded and target positions, current limits, and saturation indicators are excluded from the encoder input. Sensor and reception timestamps are used to construct causal history windows containing only measurements available at the corresponding observation time.

The motor-only input is deliberate. Measured position gives the gripper aperture and cumulative closure, velocity indicates motion direction and whether closing slows or stops, and current describes the simultaneous motor load. Their joint temporal evolution provides the information needed for the contact and phase distinctions supervised in this work. Excluding wrist video also prevents the MIM token from learning an appearance shortcut. This does not imply that telemetry replaces vision for general manipulation: camera observations remain inputs to SmolVLM, whereas MIM is restricted to encoding the physical response during gripper-object interaction.

Formally, for an observation issued at time t , MIM receives the causal sequence

$$\mathbf{X}_t = [\mathbf{x}_{t-L+1}, \dots, \mathbf{x}_t], \quad (1)$$

where each $\mathbf{x}$ concatenates normalized measured position, velocity, and current, their first-order differences, motor-status

bits, validity masks, and timing information. Samples whose acquisition or reception time exceeds t are excluded even when they occur in the same logged image interval. When a signal or temporal difference is unavailable, its validity entry is cleared so that zero-filled normalized values cannot be mistaken for a physical zero. This construction is important for deployment, where sensor packets arrive asynchronously and the policy must never depend on future telemetry.

Leader-follower demonstrations cover closing, contact, additional closing, holding, partial opening, regrasping, and release. Human reviewers annotate binary contact and five phases: IDLE, CLOSING, SQUEEZE, HOLD, and OPENING. Contact requires both fingers to visually touch the object. SQUEEZE denotes additional closing after contact, irrespective of visible deformation, whereas HOLD contains no additional opening or closing command. Video is used only to generate these targets and is not a MIM input. The MIM pretraining set contains 360 interaction episodes, with 20 episodes collected for each of the 18 objects shown in Fig. 4.

The two annotations provide complementary supervision. Contact labels identify the transition from free motion to object interaction, whereas phase labels describe how the response evolves before and after that transition. For example, both a rigid object and a compliant object may be in SQUEEZE, but they can exhibit different combinations of position advance, velocity decay, and current rise. Conversely, similar instantaneous currents may occur during impact, sustained holding, or reopening. Supervising the temporal encoder with both targets discourages it from reducing the history to a single current threshold or gripper aperture.

Measurements and first-order differences are normalized with training-split statistics and accompanied by validity masks. A learned projection with layer normalization [18] and GELU [19] maps each time step to a feature vector. A GRU [20] encodes the sequence, and an MLP maps its final hidden state to τ . Contact and phase heads supervise this shared representation, as shown in Fig. 3:

$$\mathcal{L}_{\text{MIM}} = \lambda_{\text{phase}} \mathcal{L}_{\text{phase}} + \lambda_{\text{contact}} \mathcal{L}_{\text{contact}}. \quad (2)$$

Figure 4 provides an empirical, qualitative validation of the motor evidence used by MIM. Under the standardized probing motion, higher-resistance objects show limited additional closure and stronger current response, while more compliant objects permit greater closure with a weaker current response. Objects with similar resistance occupy nearby regions even when their appearance differs. The plot therefore verifies that the measured histories contain object-dependent physical structure that a temporal encoder can exploit. MIM is not trained from these two summary coordinates; it receives the complete causal motor history and is supervised by the contact and phase objectives above.

Each term is evaluated only when its target is valid. Sampling is balanced across object families and episodes, model selection uses the validation split, and the test split is held out. Downstream modules consume τ before the two classification heads rather than their discrete predictions.

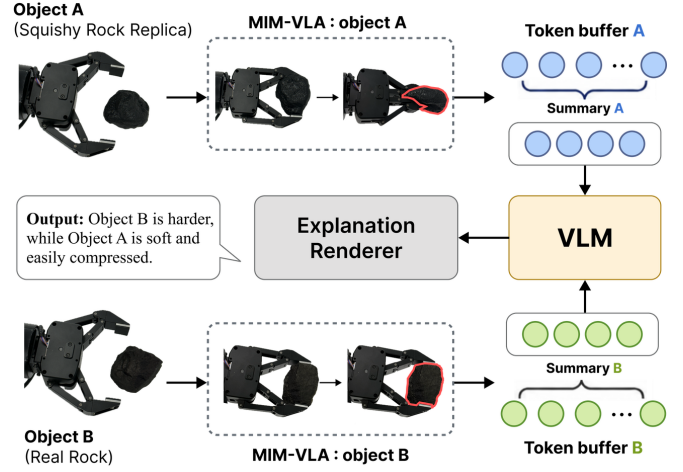


Fig. 5. Candidate-specific interaction histories are summarized independently and compared by the Material Explain Module (MEM) to produce an object choice and evidence-conditioned explanation.

Using the pre-classification latent is essential because the auxiliary classes are deliberately coarse. A phase prediction states whether the gripper is closing, squeezing, holding, or opening, but does not encode the magnitude or temporal profile of the resistance encountered within that phase. Likewise, the binary contact target cannot distinguish early fingertip contact from a stable load-bearing grasp. The continuous token preserves these within-class variations and therefore supports both low-level closure adaptation and higher-level comparison of previously explored objects. The auxiliary heads are removed from the policy loss after pretraining; they shape the representation without constraining downstream modules to make decisions from the discrete labels.

C. Interaction Memory and Material Reasoning

MIM interaction tokens are stored separately for each candidate object. For candidate $c \in \{A, B\}$, the interaction memory is defined as

$$\mathcal{B}_c = \{(\tau_i, t_i, \mathbf{m}_i, p_i)\}_{i=1}^{N_c}, \quad (3)$$

where $\tau_i \in \mathbb{R}^{128}$ denotes the MIM interaction token, t_i its timestamp, $\mathbf{m}_i$ the corresponding motor observation, and p_i the interaction phase. Each candidate memory is independently constructed during exploratory grasping. The comparison interval spans from initial contact through the subsequent closing and holding phases.

For each candidate, the MIM token sequence is combined with temporally aligned motor features and temporal information and passed to the Temporal Summary Module. A bidirectional temporal encoder processes the interaction sequence. Attention, mean, max, and last-state pooling capture complementary temporal characteristics. Each pooled context is projected to 128 dimensions, yielding the summary representation

$$\mathbf{S}_c = \{\mathbf{s}_c^k\}_{k=1}^K, \quad \mathbf{S}_c \in \mathbb{R}^{K \times 128}. \quad (4)$$

The default setting uses $K = 4$, with one summary token generated from each pooling context. Section IV-D evaluates this token capacity.

The Temporal Summary Module also predicts interaction statistics associated with position and current responses. These auxiliary objectives encourage the summary representation to preserve motor-response information relevant to candidate comparison.

As summarized in Fig. 5, the summary representations of both candidates are provided to the MEM selector VLM together with the task instruction. A learnable projection maps the summary tokens into the VLM embedding space. The VLM compares the two interaction representations and produces structured predictions for candidate selection and relative position/current responses. It also generates a natural-language explanation conditioned on the same interaction representation:

$$(\mathbf{S}_A, \mathbf{S}_B, x_{\text{task}}) \mapsto (\hat{y}_{\text{sel}}, \hat{y}_{\text{pos}}, \hat{y}_{\text{cur}}, \hat{x}_{\text{exp}}). \quad (5)$$

Training proceeds in two stages. We first freeze MIM and train the Temporal Summary Module using MIM tokens and aligned motor features. We then freeze the Temporal Summary Module and train the VLM motor-token projection, structured prediction heads, and LoRA parameters.

D. SmolVLA Integration

We retain SmolVLA’s flow-matching action formulation [2] (Fig. 2). The system figure intentionally abstracts the action block as a single expert for readability; the gripper-specific split used during fine-tuning is detailed here. A trainable projector and token-type embedding map τ into the action-expert context. A dedicated gripper expert, initialized from the pretrained action expert, receives this token and the base arm trajectory. An asymmetric attention mask allows the interaction token to attend to the original observation context but blocks the reverse path.

The base expert predicts the complete joint-action sequence. The dedicated expert predicts only the conditioned gripper channel and does not feed its output back into the base trajectory. The executed action combines base arm commands with the dedicated gripper command, preserving the original position-action interface.

Policy fine-tuning uses

$$\mathcal{L}_{\text{policy}} = \mathcal{L}_{\text{base}} + \lambda_{\text{grip}} \mathcal{L}_{\text{grip}}, \quad (6)$$

where the gripper term is evaluated only for valid MIM representations and targets. The vision-language backbone remains frozen. Both action experts, their projections, the interaction-token embedding, and the MIM encoder are updated; gripper-loss gradients reach MIM but not the original observation context. The pretrained contact and phase heads remain frozen and are excluded from the policy objective.

IV. EXPERIMENTS

A. Experimental Setup

Experiments use a 6-DoF OMY F3M follower arm with a Dynamixel two-finger gripper, an overview RGB camera, and a wrist-mounted Intel RealSense D405. Demonstrations are collected through an OMY leader–follower interface. Joint states and read-only gripper telemetry are transported through ROS 2 and Zenoh. The 100-Hz telemetry stream contains 12 command, measured-response, status, and health channels, including position, velocity, and current. Acquisition and reception timestamps ensure that each observation uses only motor samples available at that time.

The gripper-only policy used in Experiment 3 is developed from 197 policy episodes. Exporting these episodes together with 43 no-contact baselines and three high-force diagnostic negatives yields 243 episodes and 65,512 causal samples at 30 Hz with a 2-s, 200×12 motor-history window. Training uses an RTX 5090 and batch size 64. For the controlled Experiment 3 comparison, all deployed gripper-policy checkpoints are taken after 10,000 updates. The policy predicts the next absolute gripper goal position while the arm remains fixed; the Experiment 3 variants do not use camera images as policy inputs.

Table I summarizes the three real-world tasks. Experiments 1 and 2 test whether stored interaction evidence supports relative material judgments across appearance changes and under appearance ambiguity. Experiment 3 evaluates minimally compressive grasping of fragile objects.

Experiments 1 and 2 compare vanilla SmolVLA [2], SmolVLA with raw current, and MIM-VLA over 20 trials per pair. Experiment 3 compares a SmolVLA-flow gripper baseline without motor history, a temporal-current baseline inspired by the design principles of TA-VLA [9], and MIM-VLA over 20 trials per object. Because TA-VLA training code was not publicly available, the second baseline is our controlled implementation rather than an exact reproduction. All methods use identical object placement, camera observations, language instructions, and task-specific success criteria.

B. Material Evidence and Candidate Selection

Experiments 1 and 2 share the probe–compare–select protocol in Fig. 6. The robot probes the left object first and then the right object with the same exploratory grasp routine; assignment of the two candidates to the left and right positions is randomized for every trial. MIM retains tokens from first contact through additional closing and holding, and the Material Explain Module (MEM) compares equal-length summaries to select the harder candidate and generate a short explanation grounded in relative position and current responses. We collect 1,200 training episodes in total across Experiments 1 and 2 for this probe–compare–select pipeline.

We operationally define the target as the object exhibiting greater interaction resistance under this standardized probing motion. Target orderings are assigned from nominal material behavior and repeated exploratory observations, and are qualitatively checked against closure and current responses

TABLE I
REAL-WORLD EVALUATION TASKS. “UNSEEN” DENOTES HELD-OUT OBJECTS.

Exp.	Task	Evaluation protocol	Split
1	Cross-Appearance Hardness Comparison	Sequentially probe two visually different objects, select the harder object, and explain the relative response.	7 seen + 3 unseen pairs
2	Appearance-Matched Physical Disambiguation	Probe a real object and its appearance-matched soft replica, then identify the harder or genuine candidate from interaction evidence.	3 matched pairs
3	Gentle Grasping of Fragile Objects	From a fixed arm pose, detect contact and establish a stable hold with minimal compression.	8 seen + 3 unseen objects

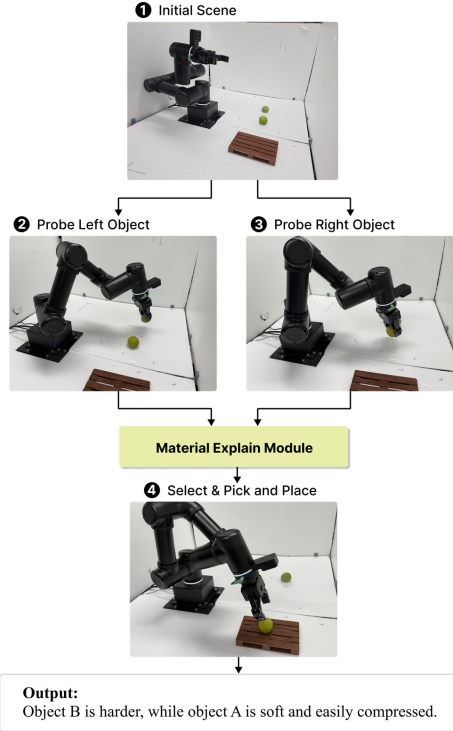


Fig. 6. Shared probe-compare-select protocol for Experiments 1 and 2. The robot observes both candidates, probes each object, compares their interaction memories, and manipulates the selected candidate.

such as those in Fig. 4. This evaluation is not intended as a calibrated measurement of intrinsic material hardness.

Experiment 1 uses visually different objects. Its seen pairs are Baseball/Sponge, Tennis Ball/Sponge, Plastic Cup/Paper Cup, Wooden Box/Toy Ball, Cardboard Box/Cylindrical Plastic Model, Toy Ball/Toy Beverage Bottle, and Wooden Box/Tennis Ball. The unseen pairs are Toy Bell Pepper/Air Blower, Golf Ball/Empty PET Bottle, and Golf Ball/Miniature Doll. Experiment 2 instead pairs a real object with a visually similar soft replica: Real Rock/Squishy Rock, Real Lemon/Lemon Squishy, and Real Banana-Milk Bottle/Squishy Replica. Selection accuracy is the primary metric for both tasks.

Table II shows that MIM-VLA achieves 75.0% aggregate selection accuracy, compared with 48.8% for vanilla SmoVLA and 29.2% for raw-current conditioning. The advantage remains on unseen cross-appearance pairs (58.3% versus 35.0% and 23.3%), while the appearance-matched task reaches 68.3%. Under this evaluation, the results are consistent

with a benefit from temporally pretrained interaction features over instantaneous raw-current conditioning; we do not interpret the comparison as a calibrated material test.

C. Gentle Grasping of Fragile Objects

Experiment 3 isolates gripper control from arm motion. The policy is trained on minimal-compression demonstrations spanning deformable and breakable objects; closing stops once the object can be held without slipping. At test time, a deterministic controller pre-opens, search-closes at low speed until MIM contact and sustained position lag agree, executes the learned hold policy for 10 s, and reopens. Additional learned closure is capped at 0.012 rad from detected contact.



Fig. 7. Qualitative evaluation of gentle grasping. Top: damage-free grasps of an unseen raw egg (left) and a seen Oreo cookie (right), each shown with its SAM 2 evaluation mask. Bottom: marshmallow deformation for MIM-VLA (left, 18.0% mask-area reduction) and the SmoVLA-flow baseline (right, 41.2%). SAM 2 masks are used only for evaluation.

Figure 7 shows representative damage-free grasps and the fixed-view SAM 2 evaluation used at maximum closure.

We record grasp success, object drop or damage, and visual deformation. A damage-free grasp requires successful pickup, a stable 10-s hold, and release with the object remaining intact; a drop, crack, rupture, or visible permanent damage at or after release is counted as failure. SAM 2 [21] is used only for evaluation: an object mask initialized before contact is propagated through the grasp clip. The camera viewpoint and arm pose are fixed across policies, the object remains unoccluded in the evaluation view, and area is measured at the maximum-closure frame during the hold. Relative mask-area

TABLE II

HARDER-TARGET SELECTION RESULTS FOR EXPERIMENTS 1 AND 2. EACH OBJECT PAIR IS EVALUATED OVER 20 TRIALS PER POLICY. BEST RESULTS ARE SHOWN IN BOLD.

Exp.	Split	Object pair	Harder target	Vanilla SmolVLA	SmolVLA + Raw Current	MIM-VLA
<i>Experiment 1: Cross-Appearance Hardness Comparison</i>						
1	Seen	Baseball / Sponge	Baseball	13/20 (65%)	9/20 (45%)	19/20 (95%)
1	Seen	Tennis Ball / Sponge	Tennis Ball	12/20 (60%)	7/20 (35%)	19/20 (95%)
1	Seen	Plastic Cup / Paper Cup	Plastic Cup	13/20 (65%)	6/20 (30%)	18/20 (90%)
1	Seen	Wooden Box / Toy Ball	Wooden Box	10/20 (50%)	9/20 (45%)	16/20 (80%)
1	Seen	Cardboard Box / Cylindrical Plastic Model	Cylindrical Model	13/20 (65%)	6/20 (30%)	18/20 (90%)
1	Seen	Toy Ball / Toy Beverage Bottle	Toy Beverage Bottle	11/20 (55%)	8/20 (40%)	14/20 (70%)
1	Seen	Wooden Box / Tennis Ball	Wooden Box	9/20 (45%)	4/20 (20%)	15/20 (75%)
1	Seen avg.	All seen pairs	—	81/140 (57.9%)	49/140 (35.0%)	119/140 (85.0%)
1	Unseen	Toy Bell Pepper / Air Blower	Toy Bell Pepper	5/20 (25%)	4/20 (20%)	11/20 (55%)
1	Unseen	Golf Ball / Empty PET Bottle	Golf Ball	8/20 (40%)	5/20 (25%)	13/20 (65%)
1	Unseen	Golf Ball / Miniature Doll	Golf Ball	8/20 (40%)	5/20 (25%)	11/20 (55%)
1	Unseen avg.	All unseen pairs	—	21/60 (35.0%)	14/60 (23.3%)	35/60 (58.3%)
<i>Experiment 2: Appearance-Matched Physical Disambiguation</i>						
2	Seen	Real Rock / Squishy Rock Replica	Real Rock	6/20 (30%)	3/20 (15%)	7/20 (35%)
2	Seen	Real Lemon / Lemon Squishy Replica	Real Lemon	10/20 (50%)	4/20 (20%)	16/20 (80%)
2	Seen	Real Banana-Milk Bottle / Squishy Replica	Real Banana-Milk Bottle	9/20 (45%)	6/20 (30%)	18/20 (90%)
2	Average	All Experiment 2 pairs	—	25/60 (41.7%)	13/60 (21.7%)	41/60 (68.3%)
Overall				127/260 (48.8%)	76/260 (29.2%)	195/260 (75.0%)

reduction is

$$R_A = 100 \frac{A_0 - A_{\text{close}}}{A_0}, \quad (7)$$

where A_0 is the pre-contact area and A_{close} is the mask area at maximum closure. For each object-model pair o , the $N = 20$ trial-level reductions $D_{o,i}$ are summarized by

$$D_o = \frac{1}{N} \sum_{i=1}^N D_{o,i}, \quad \sigma_o = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (D_{o,i} - D_o)^2}. \quad (8)$$

The fixed camera and arm pose reduce viewpoint variation. SAM 2 masks are never provided to the policy. Deformation is reported as mean $\pm$ SD; damage-free grasping is reported as successful trials over N and the corresponding rate.

Table III shows that MIM-VLA reduces mean mask-area loss on seen deformable objects from 31.4% for the SmolVLA-flow baseline and 24.0% for the TA-VLA-inspired baseline to 14.2%. On held-out deformable objects it obtains 12.5%, compared with 28.0% and 20.8%. Damage-free success reaches 50.0% on seen breakable objects and 6/20 trials on the unseen raw egg. These results are consistent with temporal motor context supporting gentler closure, while also showing substantial room for stronger fragile-object generalization.

D. Ablation Studies

This ablation is an offline evaluation of MEM using completed, held-out interaction records. It isolates nominal candidate comparison from the additional failures that can occur during real-robot probing and final action execution, and therefore is not directly comparable to the end-to-end success rates in Table II.

We vary the number K of summary tokens per candidate while keeping probing trajectories, visual observations, and training data fixed. Table IV shows that $K = 4$ yields

the best selection accuracy, explanation consistency, and swap consistency. Although $K = 2$ gives the strongest position/current F1, four tokens provide the best overall decision and explanation performance and are therefore used in all other experiments.

V. CONCLUSION

We presented MIM-VLA, an approach that uses onboard actuator telemetry instead of adding a force-torque sensor or tactile array in the evaluated setup, and converts recent gripper position, velocity, and motor-current histories into an interaction token for robot control and higher-level interaction-resistance reasoning. MIM preserves the robot’s position-control interface while adding temporal physical evidence through a gripper-specific action pathway. Across the first two experiments, MIM-VLA achieved 75.0% harder-object selection accuracy over 13 object pairs, compared with 48.8% for vanilla SmolVLA and 29.2% for raw-current conditioning. In the gentle-grasp experiment, it reduced mean SAM 2 mask-area reduction to 14.2% on seen deformable objects and 12.5% on held-out deformable objects, while improving damage-free grasp success to 50.0% on seen breakable objects and 30.0% on the unseen raw egg.

These results are consistent with the hypothesis that actuator telemetry is more useful when modeled as a temporally supervised interaction representation than when added as an instantaneous raw input. The same token interface also enables candidate-specific physical evidence to be retained for subsequent selection and explanation. Although evaluated here on pairwise resistance comparison, MIM-MEM could extend to deformability-aware sorting, tool or grasp strategy selection, regrasp after slip, and trial-based insertion. These remain future directions. Current limitations include the use of one gripper platform, the absence of calibrated force targets,

TABLE III

GENTLE-GRASP RESULTS. DEFORMATION IS MEAN $\pm$ SD OVER 20 TRIALS PER OBJECT AND POLICY; LOWER SAM 2 MASK-AREA REDUCTION IS BETTER.
BREAKABLE-OBJECT RESULTS REPORT SUCCESSFUL TRIALS AND RATES; HIGHER IS BETTER.

Split	Object	SmolVLA flow	TA-VLA-inspired	MIM-VLA (Ours)
<i>Deformable objects: SAM2 mask-area reduction (%) $\downarrow$</i>				
Seen	Marshmallow	41.2 $\pm$ 9.3	31.0 $\pm$ 7.6	18.0 $\pm$ 5.4
Seen	Empty PET Bottle	24.8 $\pm$ 6.7	19.0 $\pm$ 5.4	12.0 $\pm$ 3.8
Seen	1/3-Filled PET Bottle	17.6 $\pm$ 5.1	13.5 $\pm$ 4.0	8.5 $\pm$ 2.7
Seen	Kitchen Sponge	38.5 $\pm$ 10.2	30.0 $\pm$ 8.1	17.5 $\pm$ 5.8
Seen	Squishy Toy	35.7 $\pm$ 9.4	27.5 $\pm$ 7.3	15.5 $\pm$ 4.9
Seen	Thin Plastic Cup	30.6 $\pm$ 8.2	23.0 $\pm$ 6.2	13.5 $\pm$ 4.4
Seen macro avg.	6 objects	31.4%	24.0%	14.2%
Unseen	Paper Cup	34.0 $\pm$ 8.9	25.0 $\pm$ 7.0	15.0 $\pm$ 5.0
Unseen	Toy Soccer Ball	22.0 $\pm$ 6.0	16.5 $\pm$ 4.8	10.0 $\pm$ 3.1
Unseen macro avg.	2 objects	28.0%	20.8%	12.5%
<i>Breakable objects: damage-free grasp success (%) $\uparrow$</i>				
Seen	Cherry Tomato	4/20 (20%)	7/20 (35%)	11/20 (55%)
Seen	Oreo Cookie	2/20 (10%)	5/20 (25%)	9/20 (45%)
Seen avg.	2 objects	6/40 (15.0%)	12/40 (30.0%)	20/40 (50.0%)
Unseen	Raw Egg	1/20 (5%)	3/20 (15%)	6/20 (30%)

TABLE IV

SUMMARY-TOKEN CAPACITY ABLATION. K IS PER CANDIDATE AND TOTAL TOKENS COUNT BOTH CANDIDATES. RESULTS ARE MEAN $\pm$ SD OVER THREE SEEDS.

	Total tokens	Selection acc.	Position F1	Current F1
K	$\downarrow$	$\uparrow$	$\uparrow$	$\uparrow$
1	2	97.6 $\pm$ 0.7	78.1 $\pm$ 3.8	77.3 $\pm$ 10.3
2	4	98.1 $\pm$ 0.7	78.4 $\pm$ 1.1	81.3 $\pm$ 3.8
4	8	99.5 $\pm$ 0.7	71.4 $\pm$ 9.2	78.0 $\pm$ 3.2
8	16	96.2 $\pm$ 1.8	78.0 $\pm$ 2.4	67.1 $\pm$ 8.1

	Explanation consistency	Swap consistency
K	$\uparrow$	$\uparrow$
1	81.0 $\pm$ 13.8	91.9 $\pm$ 2.9
2	87.6 $\pm$ 1.8	90.0 $\pm$ 2.0
4	90.0 $\pm$ 2.3	93.8 $\pm$ 2.9
8	87.6 $\pm$ 1.8	89.0 $\pm$ 1.8

Each seed contains 70 test examples formed by applying both *softer* and *harder* instructions to 35 held-out candidate pairs.

sensitivity of mask-area measurements to segmentation error, and modest success on unseen breakable objects. Future work will extend the evaluation to broader grippers and objects, calibrate the relationship between motor response and force, and study how MIM-MEM memories can support multi-object, multi-step, and counterfactual physical reasoning.

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